DS203 Exercise 7: G-52

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Problem Statement

- Classify the digitized layouts of floor plan into three categories
  - Low Complexity
  - Medium Complexity
  - High Complexity
- Suggest starter designs using minimal prior information

Executive Overview

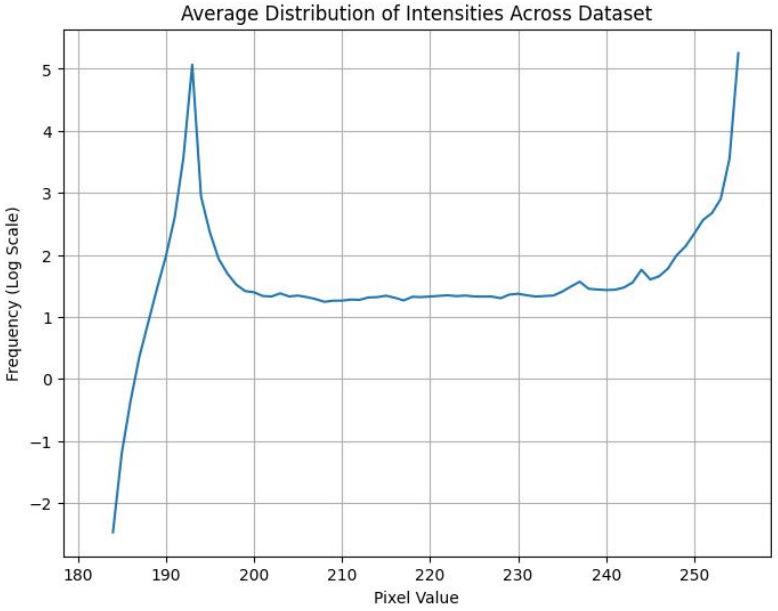
- Classifies floor plans based on complexity
  - Simple features (Unsupervised)
    - Perimeter, Area, Number of corners
  - Complex features (Supervised)
    - Bounding Box Area, Perimeter/Area
- Provides starter floor plans given bounding box area & design area
- Good balance of both machine learning & human expertise
  - Manual inspection of labels before Machine Learning
- Interpretable results
  - Easy to understand feature-output correlation
  - Aids in improving the model by feature engineering
- Low computation power thus cost effective
- Popular and trustworthy methods
  - Harris Corner Detection for feature extraction
  - K-Means Algorithm for unsupervised learning
  - Neural Networks for supervised learning

Approach Outline

- Part 1 : Classification
  - Exploratory Data Analysis
  - Feature Extraction
  - Unsupervised Machine Learning
  - Feature Engineering
  - Supervised Machine Learning
  - Experiments
  - Results
- Part 2 : Design Simplification
  - Approach
  - Results
- Challenges

Exploratory Data Analysis/Transformations

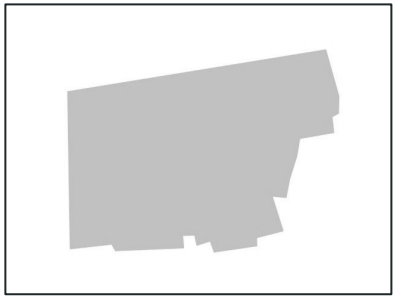
- Number of floor plans ~ 1200
- Characteristics of images
  - Image resolution : 640x480
  - Coloured dark floor plans on light background
  - Shape of floor plans : Polygons (~90%)
  - Straight lines, corners and curves
- Images **appear** to binary (see histogram on right)



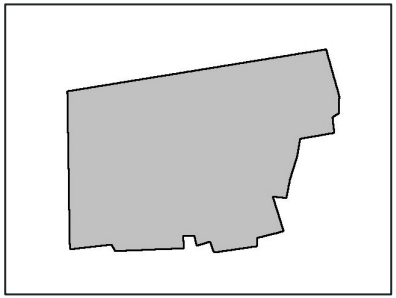
- Conversion to grayscale - yes
  - Why? No information loss occurs for given application; Easier processing
- Conversion to binary - no
  - Why? Adds noise during corner detection

Feature Extraction

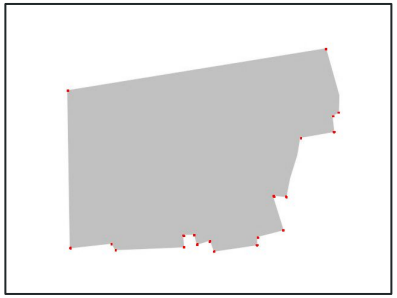
- Useful features
  - **Area** : Proportional to number of pixels below certain threshold (=225)
  - **Perimeter** : Proportional to contour length of dark pixels
  - **Bounding box area** : Minimum area rectangle around the floor plan
  - **No. of corners** : Harris Corner Detection
    - Spurious output for binary images
    - Sensitivity hyperparameter tuned to detect correct number of corners



Original Image



Perimeter

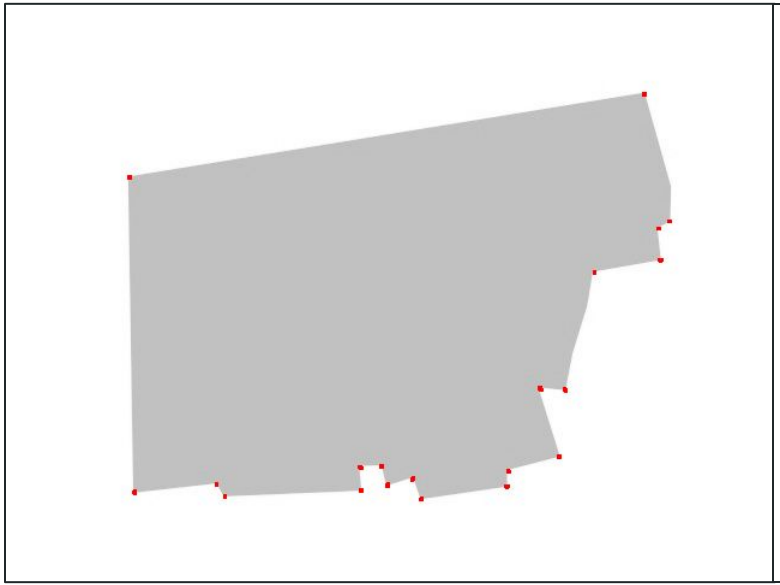


Corner Detection

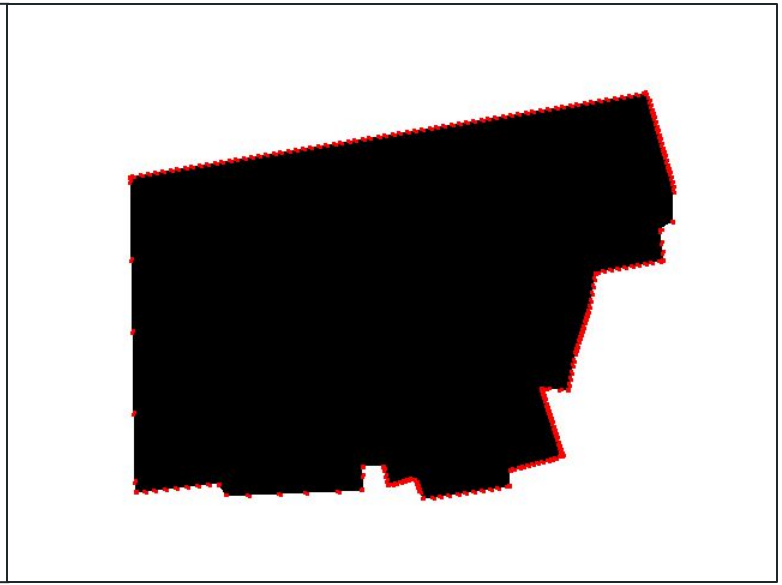


Bounding Box

Feature Extraction : Effect of Binary/Grayscale Image



Correct number of corners identified in grayscale image



Incorrect number of corners identified in binary image

Unsupervised Machine Learning for Label Generation

- K-Means Clustering Algorithm
  - Simple features for easier interpretability
    - Area
    - Perimeter
    - Number of corners
  - Required number of clusters = 3
  - Clustering interpretation
    - Low complexity : LABEL := 0
    - Medium complexity : LABEL := 2
    - High complexity : LABEL := 1
  - Manual inspection of images and generated labels to avoid misclassification

K-Means Algorithm : Input Features

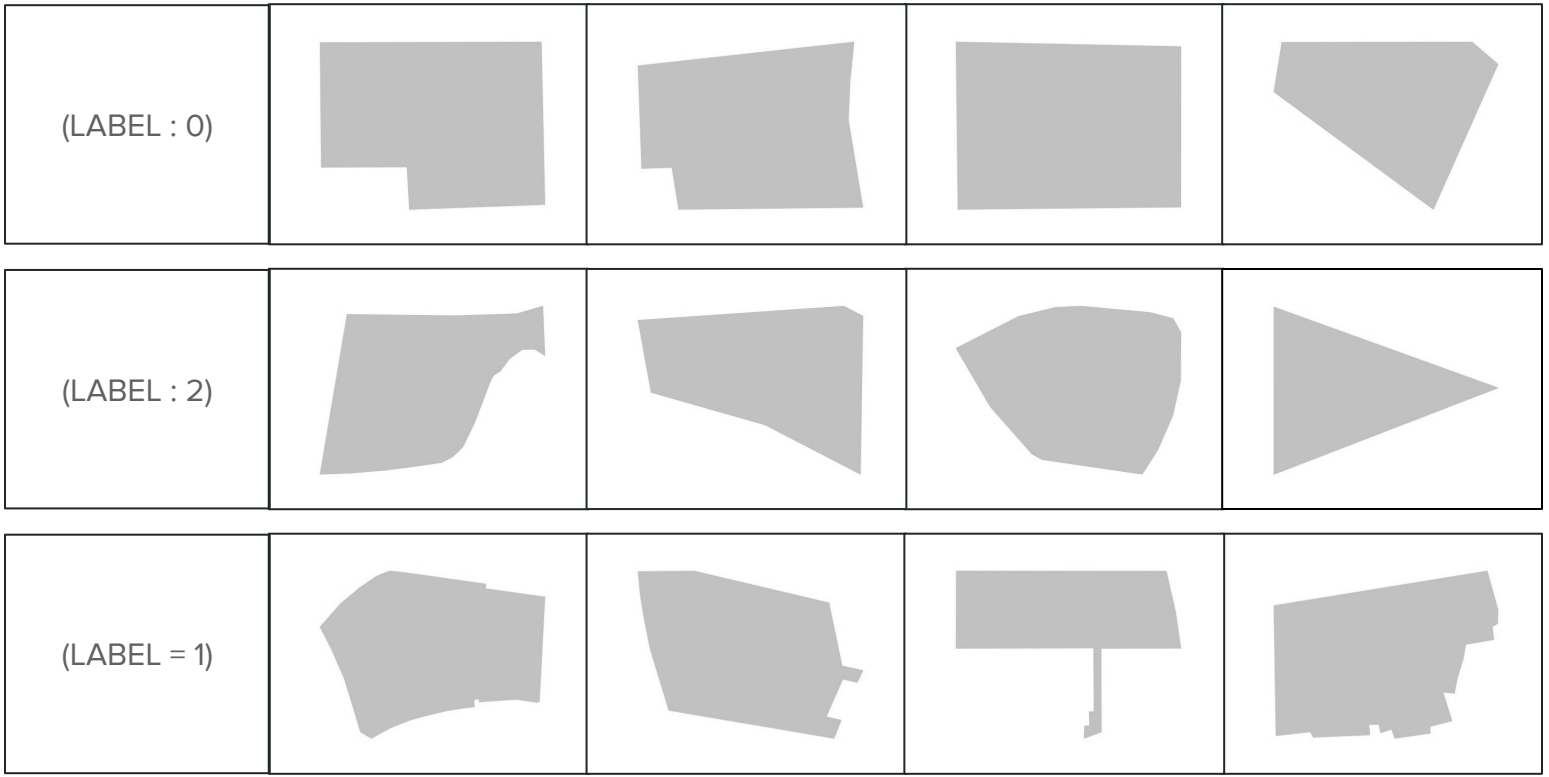
- Use simple features - Perimeter, Area and No. of Corners
  - Why simple? Output cluster should be interpretable
  - Why these features? Independant
- Features have different data range - Normalization

	Perimeter	Area	No. of corners
Mean	1496.08	122826.61	135
Standard Deviation	74.71	20664.04	65

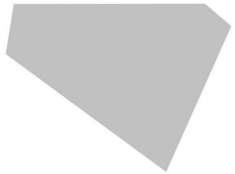
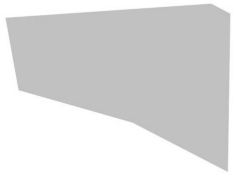
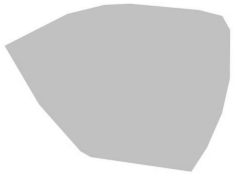
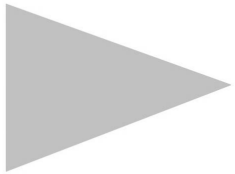
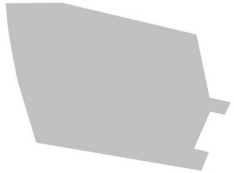
No. of corners proportional to visually visible corners

- Figure with 4 corners : 100 Harris corners
- Figure with 5 corners : 125 Harris corners and so on

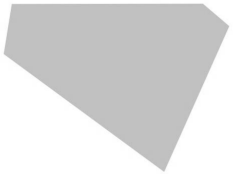
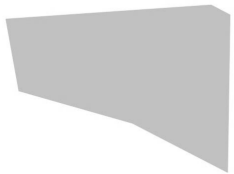
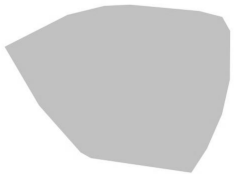
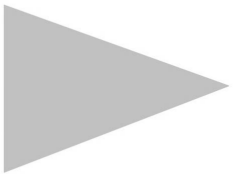
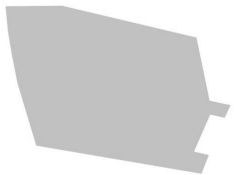
K-Means Algorithm : Outputs



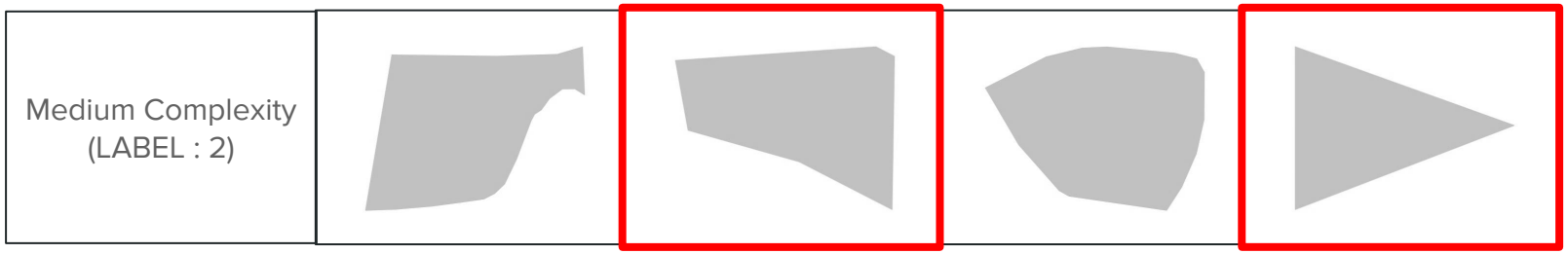
K-Means Algorithm : Interpretation

Low Complexity (LABEL : 0)					Polygons with straight lines
Medium Complexity (LABEL : 2)					Curves and straight lines
High Complexity (LABEL = 1)					Sharp protrusion and cavities

K-Means Algorithm : Misclassification

Low Complexity (LABEL : 0)					Polygons with straight lines
Medium Complexity (LABEL : 2)					Curves and straight lines
High Complexity (LABEL = 1)					Sharp protrusion and cavities

K-Means Algorithm : Label Correction



- Simple polygons with straight lines wrongly classified in Medium Complexity as opposed to Low Complexity
  - Images are noisy thus features are noisy
    - Detected corners are more/less than actually present
- Manual Inspection and label correction performed specifically for images of LABEL 2 (Medium Complexity) by reassignment

Supervised Machine Learning (SML)

- Addressing data problems
  - Features have different data ranges - Normalization
  - Apparent and absolute repeated entries
    - Visual look alike but actually not
    - Identical
- Feature engineering
  - Perimeter and area separately do not convey complete information about complexity
  - Ratio of perimeter and area more useful
    - Higher perimeter per unit area implies lower complexity
    - Lower perimeter per unit area implies higher complexity
    - Complexity(circle) > Complexity(square) (for a given area)
  - Ratio of floor area to bounding box area - measure of compactness and regularity
- Additional features
  - Area/Perimeter
  - Area(floor)/Area(Bounding Box)
- Model choice
  - Multi-layer perceptron
    - Why? Flexible architectures, Non-linear

SML Experiments - Architecture of Neural Network

- 16 different models were considered
- Model Complexity increase with number of hidden layer and size of hidden layers
  - No. of hidden layers 1, 2, 3, 4 and No. of neurons 10, 20, 30, 40
    - [10] (NN1), [10, 10] (NN2), [10, 10, 10] (NN3), [10, 10, 10, 10] (NN4)
    - [20] (NN5), [20, 20] (NN6), [20, 20, 20] (NN7), [20, 20, 20, 20] (NN8)
    - [30] (NN9), [30, 30] (NN10), [30, 30, 30] (NN11), [30, 30, 30, 30] (NN12)
    - [40] (NN13), [40, 40] (NN14), [40, 40, 40] (NN15), [40, 40, 40, 40] (NN16)
- Metrics (Average and Class wise)
    - Accuracy
    - Recall (M1)
    - F1-score (M2)
    - Precision (M3)
    - AUC (M4)

SML Observations - Architecture of Neural Network

- Accuracy, Precision, Recall, F1-score (Average, given below) for a given model are equal
  - Why? Number of false positives = Number of false negatives (Matter of chance)
- Average metrics misleading/uninformative
  - Class imbalance
    - Class 0: 63%
    - Class 1: 20%
    - Class 2: 17%
- Separate metrics calculated for each class
- Compare performance on test data to choose the best model

	NN1	NN2	NN3	NN4	NN5	NN6	NN7	NN8	NN9	NN10	NN11	NN12	NN13	NN14	NN15	NN16
Metrics (All)	0.93249	0.94093	0.96624	0.97046	0.93249	0.95359	0.96203	0.97046	0.94937	0.97468	0.97046	0.98312	0.94937	0.97468	0.97046	0.97468

Observations - Architecture of Neural Network - Class 0

	NN1	NN2	NN3	NN4	NN5	NN6	NN7	NN8	NN9	NN10	NN11	NN12	NN13	NN14	NN15	NN16
M1	0.96644	0.97987	1.00000	0.99329	0.93960	0.98658	1.00000	1.00000	0.97987	1.00000	1.00000	1.00000	0.97315	1.00000	1.00000	1.00000
M2	0.94737	0.95738	0.97705	0.98013	0.94595	0.96393	0.97068	0.98026	0.96370	0.98350	0.98026	0.98675	0.96026	0.98350	0.98026	0.98026
M3	0.92903	0.93590	0.95513	0.96732	0.95238	0.94231	0.94304	0.96129	0.94805	0.96753	0.96129	0.97386	0.94771	0.96753	0.96129	0.96129
M4	0.98833	0.99321	0.99703	0.99558	0.99199	0.99413	0.99207	0.99634	0.99085	0.99542	0.99619	0.99657	0.99115	0.99565	0.99649	0.99657

Green boxes represent best models for given metric while pink columns represent models which perform over 3 or more metrics

Recall (M1), F1-score (M2), Precision (M3), AUC (M4)

Observation - Architecture of Neural Network - Class 1

	NN1	NN2	NN3	NN4	NN5	NN6	NN7	NN8	NN9	NN10	NN11	NN12	NN13	NN14	NN15	NN16
M1	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
M2	1.00000	0.98876	0.98876	0.98876	1.00000	1.00000	1.00000	0.98876	0.98876	0.98876	0.98876	1.00000	1.00000	0.98876	0.98876	1.00000
M3	1.00000	0.97778	0.97778	0.97778	1.00000	1.00000	1.00000	0.97778	0.97778	0.97778	0.97778	1.00000	1.00000	0.97778	0.97778	1.00000
M4	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000

Green boxes represent best models for given metric while pink columns represent models which perform over 3 or more metrics

Recall (M1), F1-score (M2), Precision (M3), AUC (M4)

Observation - Architecture of Neural Network - Class 2

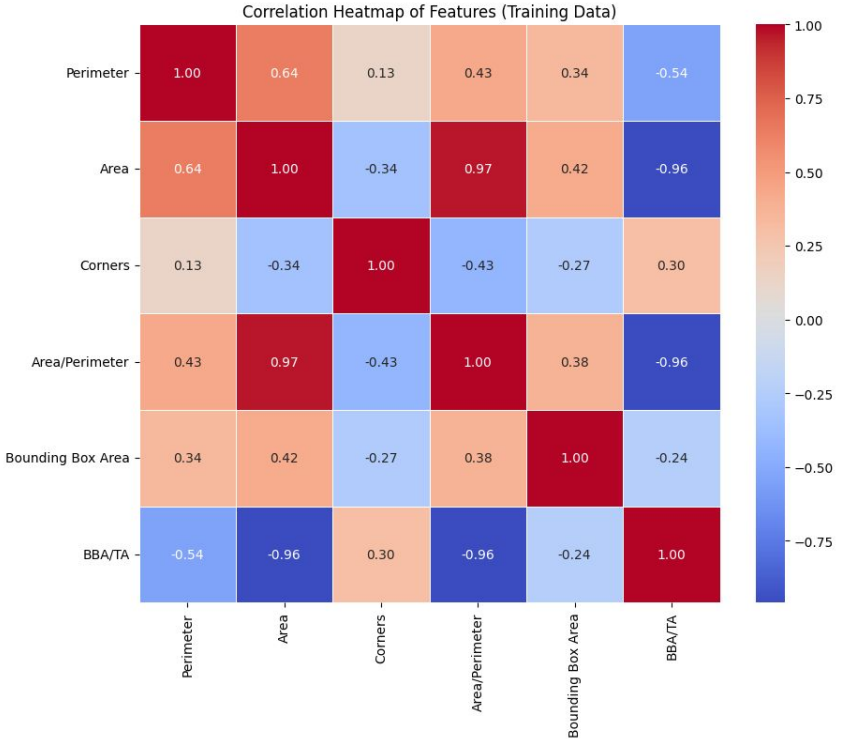
	NN1	NN2	NN3	NN4	NN5	NN6	NN7	NN8	NN9	NN10	NN11	NN12	NN13	NN14	NN15	NN16
M1	0.75000	0.75000	0.81818	0.86364	0.84091	0.79545	0.79545	0.84091	0.79545	0.86364	0.84091	0.90909	0.81818	0.86364	0.84091	0.86364
M2	0.80488	0.82500	0.90000	0.91566	0.82222	0.86420	0.88608	0.91358	0.85366	0.92683	0.91358	0.95238	0.85714	0.92683	0.91358	0.92683
M3	0.86842	0.91667	1.00000	0.97436	0.80435	0.94595	1.00000	1.00000	0.92105	1.00000	1.00000	1.00000	0.90000	1.00000	1.00000	1.00000
M4	0.97786	0.98740	0.99541	0.98905	0.98599	0.99058	0.98669	0.98976	0.98410	0.99152	0.99246	0.99282	0.98634	0.99058	0.99282	0.99399

Green boxes represent best models for given metric while pink columns represent models which perform over 3 or more metrics

Recall (M1), F1-score (M2), Precision (M3), AUC (M4)

Experiments - Feature Selection

- Correlation heat map
  - Area is highly correlated with Area/Perimeter (in dark red)
- Analyse the models
  - With Area feature (W)
  - Without Area feature (WO)



Observation - Feature Selection : Recall

	NN1	NN2	NN3	NN4	NN5	NN6	NN7	NN8	NN9	NN10	NN11	NN12	NN13	NN14	NN15	NN16
M_0_W	0.96644	0.97987	1.00000	0.99329	0.93960	0.98658	1.00000	1.00000	0.97987	1.00000	1.00000	1.00000	0.97315	1.00000	1.00000	1.00000
M_0_WO	0.93960	0.98658	1.00000	0.99329	0.97315	1.00000	1.00000	1.00000	0.97987	1.00000	1.00000	1.00000	0.97315	1.00000	1.00000	1.00000
M_1_W	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
M_1_WO	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
M_2_W	0.75000	0.75000	0.81818	0.86364	0.84091	0.79545	0.79545	0.84091	0.79545	0.86364	0.84091	0.90909	0.81818	0.86364	0.84091	0.86364
M_2_WO	0.63636	0.79545	0.75000	0.79545	0.81818	0.77273	0.90909	0.88636	0.84091	0.88636	0.93182	0.90909	0.81818	0.93182	0.93182	0.90909

Red pairs represent where model performed better WO Area feature over W Area feature while indices 0, 1, 2 on left represent classes

Observation - Feature Selection : F1-Score

	NN1	NN2	NN3	NN4	NN5	NN6	NN7	NN8	NN9	NN10	NN11	NN12	NN13	NN14	NN15	NN16
M_0_W	0.94737	0.95738	0.97705	0.98013	0.94595	0.96393	0.97068	0.98026	0.96370	0.98350	0.98026	0.98675	0.96026	0.98350	0.98026	0.98026
M_0_WO	0.91803	0.96711	0.96753	0.96732	0.96026	0.96753	0.99003	0.98350	0.96689	0.98350	0.99003	0.98675	0.96026	0.99003	0.99003	0.98675
M_1_W	1.00000	0.98876	0.98876	0.98876	1.00000	1.00000	1.00000	0.98876	0.98876	0.98876	0.98876	1.00000	1.00000	0.98876	0.98876	1.00000
M_1_WO	1.00000	0.98876	0.98876	1.00000	1.00000	1.00000	0.98876	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
M_2_W	0.80488	0.82500	0.90000	0.91566	0.82222	0.86420	0.88608	0.91358	0.85366	0.92683	0.91358	0.95238	0.85714	0.92683	0.91358	0.92683
M_2_WO	0.69136	0.86420	0.85714	0.87500	0.85714	0.87179	0.95238	0.93976	0.88095	0.93976	0.96471	0.95238	0.85714	0.96471	0.96471	0.95238

Red pairs represent where model performed better WO Area feature over W Area feature while indices 0, 1, 2 on left represent classes

Observation - Feature Selection : Precision

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
M_0_W	0.92903	0.93590	0.95513	0.96732	0.95238	0.94231	0.94304	0.96129	0.94805	0.96753	0.96129	0.97386	0.94771	0.96753	0.96129	0.96129
M_0_WO	0.91231	0.96752	0.95506	0.96807	0.97246	0.95652	0.98827	0.97865	0.97297	0.97865	0.98988	0.98829	0.97246	0.98827	0.98827	0.98829
M_1_W	1.00000	0.97778	0.97778	0.97778	1.00000	1.00000	1.00000	0.97778	0.97778	0.97778	0.97778	1.00000	1.00000	0.97778	0.97778	1.00000
M_1_WO	1.00000	0.97778	0.97778	1.00000	1.00000	1.00000	0.97778	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
M_2_W	0.86842	0.91667	1.00000	0.97436	0.80435	0.94595	1.00000	1.00000	0.92105	1.00000	1.00000	1.00000	0.90000	1.00000	1.00000	1.00000
M_2_WO	0.75676	0.94595	1.00000	0.97222	0.90000	1.00000	1.00000	1.00000	0.92500	1.00000	1.00000	1.00000	0.90000	1.00000	1.00000	1.00000

Red pairs represent where model performed better WO Area feature over W Area feature while indices 0, 1, 2 on left represent classes

Observation - Feature Selection : AUC

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
M_0_W	0.98833	0.99321	0.99703	0.99558	0.99199	0.99413	0.99207	0.99634	0.99085	0.99542	0.99619	0.99657	0.99115	0.99565	0.99649	0.99657
M_0_WO	0.98429	0.99214	0.99207	0.99230	0.98955	0.99100	0.99565	0.99626	0.99275	0.99649	0.99459	0.99565	0.99100	0.99626	0.99497	0.99619
M_1_W	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
M_1_WO	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
M_2_W	0.97786	0.98740	0.99541	0.98905	0.98599	0.99058	0.98669	0.98976	0.98410	0.99152	0.99246	0.99282	0.98634	0.99058	0.99282	0.99399
M_2_WO	0.97504	0.98610	0.98575	0.98811	0.98245	0.98316	0.99176	0.99293	0.98858	0.99458	0.99034	0.99176	0.98469	0.99305	0.99140	0.99317

Red pairs represent where model performed better WO Area feature over W Area feature and green pairs represent where model performed better W Area feature over WO Area feature while indices 0, 1, 2 on left represent classes

Results - Architecture and Feature Selection - NN12 W

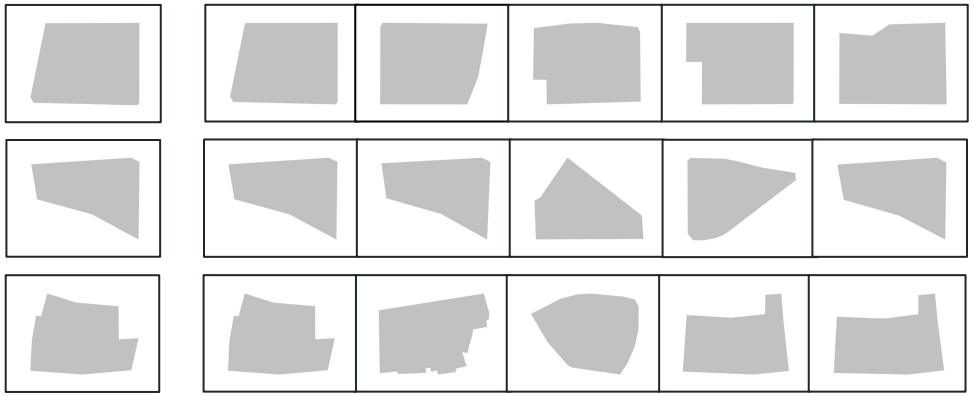
- Architecture
  - Top 4/16 models are NN10, NN12, NN14, NN16
  - NN12 performs the best among almost all the metrics and classes
  - There is no clear winner among NN10, NN14, NN16
- Feature Selection
  - Almost all the selected models perform better without the Area feature across all the metrics
  - Exception : Model NN12, NN16 for Class 0 and Class 2 for AUC
- We select model NN12 and include Area feature
  - Including Area feature performs better for Class 2 (Medium Complexity)
  - Class 2 is most difficult to identify and Area feature makes it easier

Model for Design Simplification - Approach

- Inputs : Bounding Box Area, Area and Design Complexity
- Approach
  - Select all training images with given Design Complexity
  - Find MSE of input (Bounding Box Area, Area) wrt to (Bounding Box Area, Area) of all images in training set
  - Sort MSE’s in increasing order
  - Remove identical design in dataset if present
  - Select top 15 floor plans
- User can either directly use one of the designs of modify these designs

Model for Design Simplification - Results

- Pros
  - Input (Bounding, Box Area, Area, Label)
  - Top-15 distinct designs were found (only first 5 shown)
  - Performs decently on Low Complexity and Medium Complexity Designs
  - First Design is identical to the input for all different complexities
- Cons
  - Complex designs are irrelevant (more information needed)

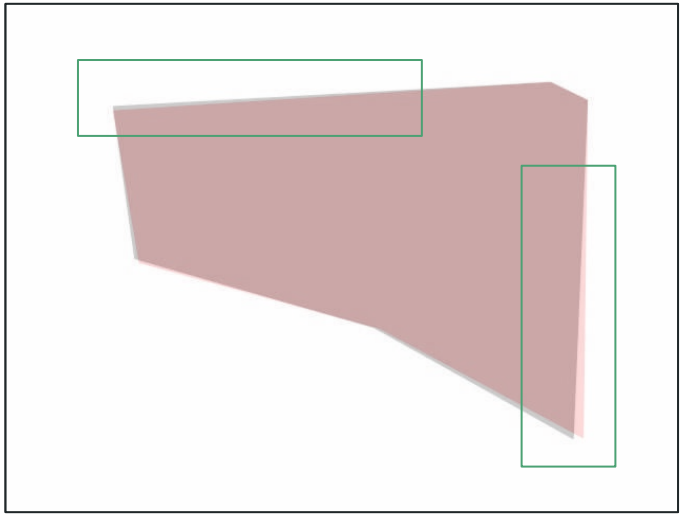


Input

Top 5 output images in order

Challenges

- No prior definition of complexity
  - Define complexity based on unsupervised learning
  - Re-assigning labels to differentiate between Medium Complexity and High Complexity difficult
- Differentiating between nearly identical and completely identical floor plans
  - Identical floor plans do not add information but useful for problem 2
  - Difficult to discard (might be slightly different) useful for problem 1
- Class Imbalance
  - Only few samples for Medium and High Complexity floor plans (20% each)
- Identifying similar High Complexity floor plans using Bounding Box Area and Area



Two visibly identical images are slightly different

Summary For Evaluation

- All the questions in the problem statement have been addressed
  - Problem 1 and Problem 2
  - Solutions are relevant and with agreement with course curriculum and requirement
  - Creative thinking
    - Combination of supervised and unsupervised methods
    - Feature engineering - Harris Corner, Perimeter/Area
- All major data science methods have been followed
    - EDA, Feature Extraction, Feature Engineering, Modelling, Analysis, Validation
- Results are backed with number of classification metrics and justifications - Recall, Accuracy, Precision, F1-score, AUC
- Everything has been neatly presented

Thank You